



Automated Data Extraction by Large Language Models: Assessing Accuracy in Comparison to Human Experts Using the Example of Visible Learning

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Abstract

By quantitatively integrating results of meta-analyses, second-order meta-analyses (SOMAs) from educational psychology address far-reaching questions of interest to adjacent scientific fields, policy, and educational practice. Conducting a SOMA, however, is time-consuming, particularly the extraction of statistical results from meta-analyses. Large Language Models (LLMs) have the potential to accelerate data extraction, as evidenced by evaluations of data extraction from primary studies. Evaluations of their accuracy in extracting results from meta-analyses to conduct SOMA are needed. We compare the accuracy of three LLMs with that of human experts extracting data from 156 educational meta-analyses investigating students' achievement, randomly chosen from the *Visible Learning* database. LLMs and humans extracted effect sizes, the number of included studies, and the number of students. To assess accuracy, we established a gold standard benchmark by resolving discrepancies between LLMs and human experts through expert adjudication. Results show that the distributions of data from all coders and the standard were similar and showed no systematic bias. Accuracy reached ICCs of 0.95/0.81 for the two human experts, and 0.96/0.97/0.96 for LLMs (Gemini2.5Pro/GPT4.1/GPTo3), with percentage agreement of 86%/80% (humans) and 81%/78%/77% (LLMs). The results demonstrate that LLMs achieve data extraction accuracy comparable to that of human experts. We discuss the conditions under which data extracted by LLMs, humans, or a hybrid of both can be considered validated for use in educational SOMAs.

Highlights

- Show a way to improve Second-Order-Meta-analyses.
- Assessing accuracy in data extraction by LLMs compared to experts.
- Experts and LLMs extracted data from 156 meta-analyses on student achievement.
- LLMs approach expert human accuracy.
- Gold standard determination from consensus calibration by multiple experts.

Extended author information available on the last page of the article

Keywords Artificial intelligence · Large language models · Meta-analyses · Systematic reviews

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Second-order meta-analyses (SOMAs) systematize and integrate meta-analytic findings, thereby functioning as a “meta-analysis of meta-analyses” (Cooper & Koenka, 2012; Schmidt & Oh, 2013). SOMAs from educational meta-analyses influence educational discussions in academia and beyond by addressing far-reaching questions of interest to the multi-disciplinary community (Emsl nder et al. 2025a; Jansen et al. 2022, 2024; Schneider & Preckel, 2017), educational practice (Higgins, 2016), and in policy (Polanin et al., 2020). The impact of SOMAs from educational studies is best exemplified by John Hattie’s Visible Learning studies (2008, 2023), which synthesized more than 2,100 meta-analyses on student achievement and sparked global debates on what works best in schools (R mer, 2019). The relevance of these methods is likely to increase further, given the growing number of SOMAs published in the field’s highest-impact journals in recent years (see Emsl nder et al., 2025b for a review of SOMAs). A hindrance to conducting educational SOMAs, as with systematic literature reviews in general, is that the method is time-consuming. For instance, the systematic retrieval of findings from full-text articles (i.e., data extraction) typically takes 125–172 min per article, with some complex extractions requiring up to 6.5 h (Gartlehner et al., 2025; Li et al., 2019). To put this in perspective, extracting data for the 156 meta-analyses examined in the present study would require approximately 390 h of specialized human labor, assuming an average of roughly two and a half hours per meta-analysis. Researchers have long sought to automate aspects of data extraction using technology (see Marshall and Wallace 2019; Thomas et al. 2025a, bc for reviews of the methodological advancements). Despite some success in accurately extracting specific variables, adoption remained low (Khalil et al., 2024) because each literature review required the development of task-specific algorithms, which demanded substantial programming expertise and offset potential time savings (Marshall & Wallace, 2019). Recently, developments in artificial intelligence, particularly large language models (LLMs; popularized through applications such as ChatGPT and Google’s Gemini), promise substantial time savings (e.g., reproducing 12 systematic reviews in two days; Cao et al., 2025; 25% time saving for a new review, Wang et al. 2025b), without requiring programming expertise (Gartlehner et al., 2025). LLMs are algorithms trained on massive-scale text datasets, enabling them to follow instructions (Blau et al., 2024), including those aimed at extracting data from study reports and manuscripts (Gartlehner et al., 2024a; Jansen et al., 2025), across contexts. Recognizing the potential large-scale application of LLMs in meta-analytic research, the leading synthesis organizations Cochrane and Campbell have recently issued guidelines for the responsible use of AI (Thomas et al. 2025a, b, c). While they encourage the use of LLMs under meaningful human control (Siebert et al., 2023), the RAISE working group (Responsible Use of AI in Evidence Synthesis)

states that evaluation studies should determine whether an AI performs adequately in a given context (Flemyng et al., 2025).

This requirement is currently unmet for the use of LLMs in educational SOMAs. Prior evaluation studies focused on extracting findings from primary studies (e.g., Konet et al., 2024; Reason et al., 2024), which report results differently than meta-analyses. For instance, whereas primary studies report information for a single study, meta-analyses report results for many included studies alongside one or more integrative summary effects, requiring LLMs to distinguish between study-level and aggregated results and to identify which estimates are relevant for second-order synthesis. Furthermore, prior approaches in evaluation studies impose limitations: they compared LLM-extracted and human-extracted data, treating the humans as error-free, and counted all differences between LLM and humans as errors made by the LLM (see Jansen et al., 2025, for a discussion). Yet, even human experts make errors (Lombard et al., 2002; Mathes et al., 2017). Therefore, current research does not provide a reliable basis for comparing LLMs with the current standard.

We aim to evaluate the use of LLMs for data extraction in educational SOMAs by assessing the accuracy of three state-of-the-art LLMs in extracting key variables (effect sizes, study counts, participant counts) from 156 educational meta-analyses. This enables educational researchers to synthesize evidence across meta-analyses more frequently, more transparently, and at a larger scale than is currently feasible.

Background

Second Order Meta-Analyses

SOMA has a thirty-year history in education (Sipe & Curlette, 1996), and can be thought of as conducting a meta-analysis of meta-analyses (also called “umbrella reviews”, Fernandez et al., 2025; “supersyntheses”, or “meta-meta-analyses,” Levin, 1993). Researchers conducting SOMA follow the same steps as those performing a meta-analysis of empirical studies, with the difference that the unit of analysis is not empirical studies but meta-analyses (Sipe & Curlette, 1996; see also Schmidt & Oh, 2013, for methodological differences). These steps within SOMA include reviewing all meta-analyses on a specific topic, systematically extracting relevant findings from these meta-analyses (called data extraction or coding), and quantitatively synthesizing these extracted findings (see Schmidt & Oh, 2013, for statistical methods). The results of SOMAs typically include an average effect size, a distribution of effect sizes, and the potential to explain heterogeneity through moderator analyses.

SOMAs have multifaceted utility in improving education for scientific (see Gurevitch et al., 2018) and political communities (e.g., McKinnon et al., 2015). First, they enable systematic comparisons of meta-analytic effect sizes, thereby revealing inconsistencies or potential errors across meta-analyses and explaining differences between results (Van Bavel et al., 2016). Second, SOMAs can support meta-analytic results by calculating mean effect sizes and conducting heterogeneity tests. Confirming meta-analytic results is needed, as meta-analyses are highly influential within the research community (Alexander, 2020; Schneider & Preckel, 2017). Third, SOMAs

results indicate the generalizability and limitations of studies in the field. Fourth, SOMAs can inform debates about which effect sizes are small, medium, or large by providing the distribution of effect sizes in a given field (Funder & Ozer, 2019; Kraft, 2020).

To date, the largest, most prominent, and impactful application of SOMA of educational studies is John Hattie's Visible Learning project (2008, 2023). This landmark work synthesized findings from over 2,100 meta-analyses summarizing all influences on students' achievement. For each meta-analysis, the website Visible Learning Meta-X (<https://www.visiblelearningmetax.com>) provides the effect size, the number of included studies, and the number of students included. Building on Visible Learning, the utility of SOMAs in education is evident by their growing application across diverse domains, including academic motivation and self-concept (Jansen et al., 2022, 2024), teacher-student relationships (Emsländer et al., 2025a), parental involvement (Kim, 2022), problem- and inquiry-based learning (Koçoğlu & Kanadlı, 2025; Öztürk et al., 2022), and instructional technologies (Bernard et al., 2014; Hew et al., 2021; Martin et al., 2022; Tamim et al., 2011). These syntheses typically integrate a substantial volume of literature, with the number of included meta-analyses ranging between 12 and 2600 (Median=22) and the number of extracted findings per meta-analyses up to 56 (Jansen et al., 2022).

Extracting Data for SOMA

The utility and quality of SOMA rely on the quality of data extraction (Lakens et al., 2016). Data extraction is the process of translating information from scientific articles (i.e., the individual meta-analyses comprising the sample of a SOMA) into a structured format, such as a table, to enable systematic comparisons and syntheses of empirical findings (Glass, 1981; Lipsey & Wilson, 2001). Data extraction processes are challenging, leading to substantial variation in quality (see Schmidt et al., 2025, for a review), ranging from error rates of 40% (Lombard et al., 2002; Mathes et al., 2017) to perfect extractions (Belur et al., 2021; Lombard et al., 2002).

Guidelines for data extraction (Higgins et al., 2023; APA, 2020) propose that each piece of information be extracted by at least two coders with extraction expertise (Orwin & Vevea, 2009), allowing errors to be identified and the reasons for discrepancies to be discussed. The extraction accuracy between the coders should be evaluated using multiple metrics that assess different aspects of rating quality (e.g., percentage exact agreement between extractions, intra-class correlation, systematic bias). To establish a final value for the analyses, the discrepancies should be resolved through discussion or arbitration (APA, 2020).

Following these guidelines requires substantial time and financial resources spent on data extraction. Extracting data from a single article takes an average of between two (Gartlehner et al., 2025) and three hours (Li et al., 2019). As a result, data extraction from all included articles consumes the majority of resources available in the lengthy and costly process of conducting systematic reviews (Mutinda et al., 2022) that takes approximately 67.3 weeks from initiation to publication (Belur et al., 2021; see also Beller et al., 2013) and up to \$100,000 (USD) in financial resources (Borah et al., 2017; Michelson & Reuter, 2019). Applying standard estimates for manual data

extraction, a median-sized SOMA requires approximately 44 to 66 h of expert labor, whereas larger reviews (e.g., Jansen et al., 2022) require more than 300 h.

Accelerating Data Extraction Through Technology

For the last 20 years, researchers have been trying to accelerate the process of data extraction by using artificial intelligence (see Marshall & Wallace, 2019, for an introduction to the topic; Jonnalagadda et al., 2015; van Dinter et al., 2021; Schmidt et al., 2021, for reviews documenting successive methodological advances; and Wagner et al., 2021, for a critical discussion). We use the term artificial intelligence (AI), similar to the group of *Responsible use of AI in evidence Synthesis* (RAISE; Thomas et al. 2025a, b). AI refers to technologies that enable machines to effectively perform highly complex tasks that would otherwise require human intelligence if a person were to perform them.

Traditionally, the best performance was achieved by developing specific AI algorithms for each literature review, which yielded near-human accuracy in data extraction from empirical studies (Mutinda et al., 2022) and made the process more efficient (Li et al., 2019; Saldanha et al., 2016). Yet, adoption was limited across fields (see International Collaboration for Automation of Systematic Reviews; O'Connor et al., 2019) and was especially low in the social and behavioral sciences (Khalil et al., 2024; Legate et al., 2024).

A hindrance to adoption was that researchers had to develop new algorithms for each literature review. These developments required specialized training data, technical expertise, and manual preprocessing of articles (Marshall & Wallace, 2019), thereby eroding the efficacy gains that could be achieved with AI. This barrier is pronounced for SOMAs, whose reporting structures differ markedly from the highly standardized formats common in primary medical studies, for which many extraction methods were originally developed. Whereas primary studies typically report a single set of results, SOMAs summarize information across multiple studies and additionally report cumulative effect estimates, thereby introducing distinct and more complex failure modes for automated systems. A further practical reason hindering adoption is that traditional algorithms did not evolve into widely available, end-to-end tools that allow researchers to run full-text extraction at scale with minimal setup.

Advances in AI, particularly the rise of Large Language Models (LLMs), popularized through applications such as ChatGPT, offer a potential solution to the need for specialized algorithms (Hill et al., 2023) and raise the potential adoption of automated data extraction. Unlike previous AI models, LLMs can follow human instructions across multiple contexts in natural language because they were trained on vast amounts of diverse data in natural languages (Kaplan et al., 2020). Hence, researchers can more easily include their extraction rules from their codebooks into the algorithms than previously. At the same time, accessible LLM-based research assistants are diffusing rapidly, offering workflows for uploading documents and exporting data tables with minimal configuration (e.g., Helms Andersen et al., 2025). This potential of LLMs to advance the processes within systematic reviews has been widely acknowledged, for instance, in the Cochrane Handbook (Higgins et al., 2023) and the scientific community (e.g., Bail, 2024; Qureshi et al., 2023; Riaz et al.,

2024; see Khraisha et al., 2024, for an empirical review of the qualities of generative AI for data extraction) sparking discussion on how to use AI data extraction while maintaining meaningful human control.

Meaningful Human Control is a widely discussed construct in AI ethics and describes the capacity of a human–AI system to remain responsibility-preserving, such that human agents can meaningfully influence system behavior and be appropriately accountable for its outcomes (e.g., Amoroso & Tamburrini, 2020; Siebert et al., 2023). For the case of SOMAs, that means that researchers can use LLMs and still be able to identify errors and influence the output. Establishing the control implies a staged evaluation. First, establish a baseline accuracy to assess predictability and basic viability. Second, compare against the status quo (expert human coders) to determine whether LLM usage can improve existing practice. Third, evaluate responsible integration patterns (e.g., LLM as an independent second coder: Jensen et al., 2025; Reason et al., 2024; LLM as a judge: Cao et al., 2025; LLM as an assistant to a human coder: Gartlehner et al., 2025) to balance speed and quality.

The current state of research satisfies the first stage in some contexts by demonstrating that LLM extraction can achieve high accuracy (Clark et al., 2025; Hu & Geng, 2024; Jansen et al., 2025; Schmidt et al., 2025). Those studies used published meta-analyses and re-conducted data extraction with LLMs to compare human- and LLM-extracted data. The evidence stems mostly from the field of medicine (Caponio et al., 2025; Clark et al., 2025; Hu & Geng, 2024; Schmidt et al., 2025), but also includes some psychological studies, including educational (Jansen et al., 2025).

Comparisons against the status quo are missing because studies treated the human-extracted data as an error-free gold standard. That means, any discrepancy between the human- and LLM-extracted data is assigned as an error from the LLM by default. While understandable from a pragmatic standpoint, treating human-extracted data as perfect overlooks the inherent fallibility of human data extraction (e.g., Mathes et al., 2017). Hence, AI is compared to a standard of absolute perfection against a benchmark that is itself imperfect. What is needed is a study comparing LLM data extraction accuracy to experts' accuracy as a measure of the status quo.

The Present Study

SOMAs from educational meta-analyses are impactful, yet their production is currently hindered by the labor-intensive and error-prone nature of manual data extraction. To accelerate the process using LLMs while maintaining meaningful human control, we provide evidence on how accurately LLMs extract data from educational meta-analyses. We investigate two research questions to address whether LLMs already fulfill the first two stages of meaningful human control:

1. *What is the accuracy of current large language models when extracting (a) effect sizes, (b) the number of included studies, and (c) the number of participants from educational meta-analyses?*
2. *How does their accuracy compare to that of expert human coders, representing the current status quo?*

To address the first research question, we use the standard approach in the literature (e.g., Jansen et al., 2025) and compare LLM extraction to the correct ‘gold standard’ extractions. The results will allow researchers conducting SOMA on educational meta-analyses to assess how much of the findings from other contexts can be generalized to their use case. To address RQ2, we introduce a methodological leap by comparing LLMs’ accuracy against the actual variability found in expert practice. Results will inform researchers conducting SOMA if using LLMs will likely increase or decrease their overall extraction accuracy.

We assume for RQ 1 that for three variables (effect sizes, study counts, participant counts) LLMs will meet conservative accuracy thresholds against adjudicated references (for effect-sizes: mean absolute error (MAE) ≤ 0.05 , intraclass correlation (ICC) ≥ 0.90 ; for the counts percentage agreement ≥ 0.75), based on the evidence that recent LLMs handle structured, explicitly stated data (Jansen et al., 2025). For the status-quo comparison (RQ2), we specify non-inferiority to expert coders with pre-set margins (agreement within -5% points, MAE within $+0.02$, ICC within -0.05). We also predict no systematic bias (i.e., no mean bias towards too high or too low numbers measured by signed error in Bland–Altman plot). Together, support for our hypotheses would provide the empirical foundation for applying LLMs to SOMA data extraction.

Method

Transparency and Openness

All data, the LLM prompts, and analysis scripts are available in the Supplementary Material at <https://osf.io/q5n7f>. This study was not preregistered. During the preparation of this work, the authors used GPT–5 and Gemini 2.5; 3.0 Pro to generate ideas and improve language. Any use of generative AI in this manuscript adhered to ethical guidelines for the use and acknowledgment of generative AI in academic research (Porsdam-Mann et al., 2024).

Procedure

Overview

We address the research questions by systematically evaluating the accuracy of LLMs’ and human experts’ data extraction for the variables required to conduct SOMA (effect sizes, number of studies, and number of participants) across 156 meta-analyses of students’ achievement. The meta-analyses were randomly selected from the Visible Learning database, which is a comprehensive review including potentially all meta-analyses on students’ achievement (for more details, see Hattie, 2008, 2023; Hattie & Timperley, 2007). The LLMs were state-of-the-art models at the time of data collection (e.g., Gemini 2.5Pro/GPT 4.1/GPT-o3). Human-expert data were obtained from the Visible Learning database and independently extracted for this article by the first and last authors.

Data extraction accuracy was assessed by comparing data extracted by LLMs and human experts with two reference standards (silver and gold). The silver standard was established by a single author, who verified the LLM and expert human codings against the data from the original publication. The gold standard was established through joint discussion between the first author, the last author, and John Hattie for all cases in which the silver standard diverged from Visible Learning or where uncertainty remained in the extraction process. A schematic illustration of the study procedure is shown in Fig. 1. The figure summarizes the steps taken in the three phases of the study procedure. Detailed descriptions of how to replicate each step are provided below.

Phase 1: Preparation of Data-Extraction

Phase 1 specifies the inputs and rules needed to run the extraction. Our study focus on automatically extracting data for a SOMA on students' achievement because students' achievement is of high relevance to educational research (e.g., Köller, 2012), and the literature provides human expert ratings for those meta-analyses. The variables needed to conduct SOMA (i.e., effect size, number of included studies, and participants; Schmidt & Oh, 2013) have already been extracted and are transparently reported in the Visible Learning database (Hattie, 2023).

Sampling and Selection of Meta-analyses From the meta-analyses included in Visible Learning, we randomly selected a sample of 200 for inclusion in the study. The randomization was performed by assigning a random number to each meta-analysis in Visible Learning, sorting the entries in ascending order by that number, and selecting the first 200 entries. After collecting the full-text articles in PDF format and reading them, 156 meta-analyses were included in the data extraction (see the Data Table in the Supplementary Material for references). The reduction of 44 studies resulted from applying several criteria. Specifically, 23 meta-analyses were excluded because the full text could not be located or accessed (most were abstracts from conferences). An additional 12 were removed because they were in non-machine-readable formats (e.g., scanned images) incompatible with the LLM text-processing requirements at the time of data collection. Three meta-analyses were excluded as they were not in English or German; this scope was defined by the linguistic competence of the expert human coders to ensure the validity of the extraction. Four meta-analyses were excluded because the Visible Learning database listed multiple effect sizes for a single publication based on its own internal categorization framework, a decision not derivable from the source text alone and thus outside the scope of an autonomous extraction task. Two articles were excluded because they were not meta-analyses (and they have since been removed from MetaX).

Developing Codebook We iteratively developed a codebook that specifies how LLMs and human experts should perform data extraction from the meta-analyses on students' achievement. The initial codebook followed the specification within Visible Learning: (1) The effect size was defined as the most overarching effect size in the metric Cohen's d , describing the relation between the focal variable and students'

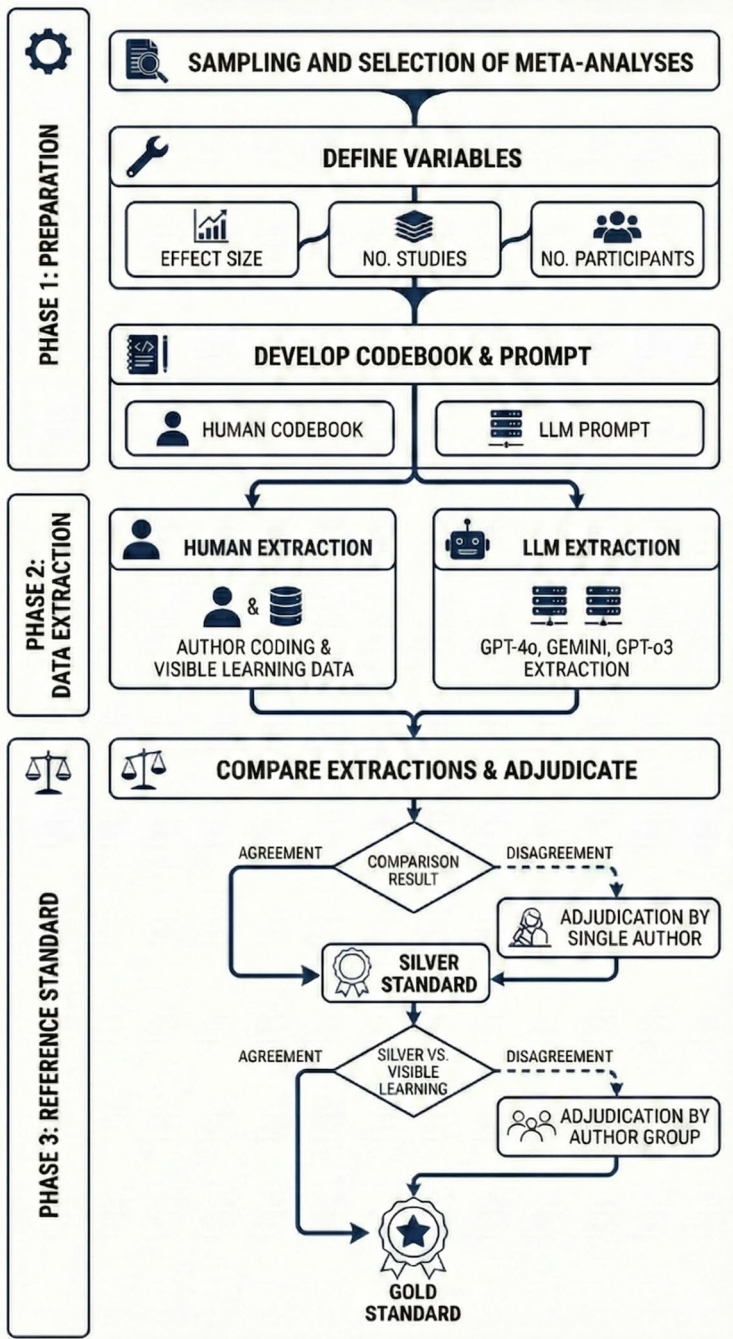


Fig. 1 Schematic Illustration of the Study Procedure

achievement. In cases with multiple effect sizes from different analyses about the same dataset, we extracted the authors' reported value as the final, most statistically robust result (i.e., after outlier and sensitivity analyses). If multiple summary effects were reported across different datasets (e.g., across different domains), the effect sizes were averaged using a simple mean. We converted alternative metrics (e.g., r) to Cohen's d and rounded all values to two decimal places. (2) The number of included primary studies referred to the number of studies contributing to the specific extracted summary effect. (3) The total number of participants referred to the aggregated sample size of the primary studies contributing to the extracted summary effect. If the number of studies or participants was not explicitly reported, it was calculated when all necessary information (e.g., participant counts in subgroups or from tables) was available in the article.

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We independently tested the codebook on a pilot sample of ten meta-analyses not included in the main sample to get a shared understanding of the coding. We drew the pilot sample of meta-analyses from the field of educational technology (see Supplementary Material for the list of pilot meta-analyses). Familiarity with the content is recommended for data extraction (Orwin & Vevea, 2009), and we had particular expertise in that context (Jansen et al., 2024b; Jansen et al., 2025b). To extract data from the pilot sample, the authors used only the PDFs and the codebook. Based on the pilot coding, the authors iteratively refined the codebook by discussing discrepancies and systematically integrating explicit rules for handling edge cases. Examples of edge cases were how effect sizes for multiple achievement measures or domains were integrated. We used the simple mean for integration to yield results comparable to those in Visible Learning. Another discussion point concerned how to report results from fixed-effects and random-effects models. While random-effects models are more appropriate in educational research because they account for the large between-study heterogeneity, fixed-effect models increase comparability between studies, as

most meta-analytic results (especially from moderator analyses or older studies) are calculated under fixed-effect assumptions. We decided to extract the results from the random-effects models.

Prompt engineering The prompt engineering for LLM data extraction employed a multi-step methodology (consistent with recommendations from Murton et al., 2025). It began with a structured initial prompt based on the codebook, followed by automated refinement, and concluded with iterative manual testing and improvement. The initial prompt was structured around a template proven effective for reproducible data extraction (Hu et al., 2024). This template consisted of five distinct segments to guide the LLM's task, separated by delimiters: < study>, < context>, < task>, < procedure>, and < formatting> (see Supplementary Material A). The < context> segment grounded the model in its role, instructing it to act as a research assistant, extracting data for meta-analyses on academic achievement in support of a large-scale replication of Hattie's Visible Learning study. The < task> segment explicitly delineated the required data points and the logic for their extraction. It instructed the model to first identify the authors and provide a two-sentence summary, then extract the specific variables: the factor affecting achievement, the overall effect size, the number of studies (k), the number of effect sizes, and the number of participants (N). The < procedure> segment guided the model's search strategy, directing it to scan the abstract, results, and methodology sections, with a particular focus on summary tables. Finally, the < formatting> segment enforced a standardized JSON output with predefined keys (effect sizes, studies, participants) to ensure all extractions were machine-readable and consistently structured. Following the initial design, the prompt underwent a refinement phase (see Helms Andersen et al., 2025; Mahmoudi et al., 2024 for a similar approach; Thomas et al. 2025a, b, for methodological recommendations) to operationalize the extraction rules in a format the model can follow (i.e., to translate domain knowledge and decision rules into explicit instructions and a constrained output schema). In the refinement phase, we began optimizing the prompt for Gemini 2.5 Pro (see Supplementary Material for the chats), as LLMs have been shown to be effective prompt engineers (Yang et al., 2023). Then, we used the machine-optimized prompt to extract data from the same ten meta-analyses we used in the pilot with the human experts (see Supplementary Material for the initial outputs). These training meta-analyses were strictly separate from the later test set to avoid data contamination (Thomas et al. 2025a, b). We analyzed the extraction and inference processes of the models (i.e., the "reasoning" of GPTo3 and Gemini 2.5 Pro) to identify where they diverged from expert conventions. To resolve these discrepancies, we implemented a decision-making hierarchy through three specific priority rules in the prompt: (1) Initial iterations revealed that models occasionally extracted effect sizes from specific moderator or subgroup analyses (e.g., gender-specific results) rather than the study's primary synthesis. We therefore required the model to prioritize values explicitly labeled as "overall," "mean," or "combined," ensuring alignment with the focal outcome domain described in the abstract. (2) In cases where reporting within a meta-analysis was inconsistent, which were mostly discrepancies between the numerical values in the abstract and the results section, we established a priority rule instructing the model to extract the findings from the

abstract, to align with the original process of building the Visible Learning database. (3) Models were observed to extract effect sizes based on preliminary samples before outlier removal. We instructed the models to prioritize the final, most robust estimate (e.g., post-outlier removal) when presented as the study's main finding. We continued the refinement process in three iterative rounds until the rules were formulated in a way to consistently extract the target value. The percentage agreement was ~80% in the pilot set (see Supplementary Material for the extracted data and comparison to the standard). The final prompts are available in the Supplementary Material.

By the end of the preparation phase, we had defined a sample of 156 meta-analyses, collected the source PDFs, and established a validated codebook for both human and LLM coders to ensure a shared understanding of variables and procedural hierarchies. No further rule changes were made during the extraction of the 156 meta-analyses. Phase 2 describes how human coders and LLMs applied these frozen rules to the PDFs, how outputs were recorded in machine-readable form, and how discrepancies were flagged and adjudicated.

Phase 2: Data-Extraction

Data extraction by Human-experts The second phase aimed to collect data extracted by LLMs and human experts. All data were entered into a Microsoft Excel file, with one row per meta-analysis. For each coder, we created three columns, one for each extracted variable. We obtained the first human-expert dataset from the Visible Learning database (queried from Hattie in April 2025; this version was released as MetaX 1.6 in February 2026).

Additionally, we conducted a second, independent extraction, henceforth referred to as author coding. For author coding, the 156 meta-analyses were randomly divided, with the first and last author independently coding 78 meta-analyses in June 2025 using a standard PDF reader and Microsoft Excel, without any AI support or access to the Visible Learning database. The authors were also considered experts because they had extensive experience with meta-analyses in education (e.g., Südkamp et al., 2012; Möller et al., 2020; Meyer et al., 2023; Jansen et al., 2024) and due to their in-depth familiarity with the Visible Learning research (fulfilling the recommendations of Orwin & Vevea, 2009). John Hattie was not involved in this stage.

Data extraction by LLMs For the LLM data extraction, we selected Gemini 2.5 Pro, GPT 4.1 (gpt-4.1-2025.1-04-14), and GPT o3 (o3-2025-04-16) for data extraction using two criteria: (a) the ability to process the entire PDF within a $\geq 64k$ -token context window (to prevent article truncation and task-splitting). Models with smaller contexts require preprocessing, which reduce the practical usability and efficiency (Konet et al., 2024) of automated data extraction methods, and (b) top-tier placement in the Chatbot Arena as of April 2025 as a proxy for pairwise user preference and perceived output quality (Chiang et al., 2024).

We have chosen not to compare different prompting styles because their value changes drastically over time, making any findings unstable. For instance, while chain of

thought prompting has been effective for the LLMs under study (Mahmoudi et al., 2024), the introduction of models optimized to enhance test-time compute (such as the o-series from OpenAI; Jaech et al., 2024) makes chain of thought prompting detrimental (Meincke et al., 2025). Further, we decided against using open-weight models because, although they provide the flexibility needed for in-depth methodological investigations and improvements (Widder et al., 2024), closed-weight models are currently more powerful in terms of data extraction (e.g., Jansen et al., 2025).

To automate LLM-based and data extraction across all primary studies, we developed a Python application that interacted with the LLMs' APIs. The code is available in Supplementary Material C. The application looped the query into the LLM, processing each meta-analysis separately with PyPDF2 and submitting the corresponding prompt via the API. In other words, all three variables were extracted within a single full-text using an independent (stateless) API request per study, to avoid potential carryover effects that can arise when an LLM processes multiple study texts within the same conversational context (Hankache et al., 2025). The model's hyperparameters were set to their default values. Upon completing all queries, the application generated a table listing the corresponding LLM responses for each variable and each PDF file processed.

The total computing fees were \$50 to extract the three variables from 156 PDFs using three LLMs (GPT 4.1 = \$ 8, Gemini 2.5 Pro 2 = \$2, o3 = \$40). By contrast, employing human research assistants would have cost about \$26,832 (assuming ~3 h per study [Li et al., 2019] at an hourly rate of \$20), and the actual resources spent were higher due to the salaries of the more senior authors. Automated extraction of the 156 meta-analyses took less than 6 min per LLM.

We manually checked whether the LLM outputs were in the expected format. While most outputs adhered to the desired format, some included additional text. We employed a Python script to clean the data by extracting only the JSON specified in the prompts. We then imported these cleaned JSON files into the master Excel workbook and mapped the JSON keys to the corresponding variable columns. After import, we performed a manual sanity check to catch formatting issues that can survive automated parsing (e.g., leading zeros, commas vs. decimal points, or fields shifted due to missing values). Cleaning took approximately one hour per LLM. The data extractions were conducted between April 10 and April 20, 2025.

Phase 3: Developing Reference Standards

To calculate accuracy, we compared the extractions from each human and LLM source against two rigorously developed benchmarks: a silver standard and a gold standard. Those standards representing the best estimates of the "correct" value for the data extraction.

Silver standard We established a "silver standard" benchmark by triangulating results from the three primary sources: the Visible Learning database, independent author coding, and the Gemini 2.5 Pro extractions (which showed the highest initial LLM accuracy). Data points were considered validated if the two human experts and the LLM produced identical values. For each data point with discrepancies across these

sources, one author re-examined the study's full text alongside the extracted values according to the codebook. Adjudication was conducted by either the first or the last author, with each meta-analysis assigned to the author who had not completed the initial coding. This procedure reduced the risk of self-confirmation bias in favor of one's own earlier coding. The result of an adjudication could be the confirmation of a prior extraction or the determination of a new value that differed from all previously coded entries (15 cases). We adjudicated discrepancies within 101 meta-analyses. No adjudication was conducted for the remaining meta-analyses because all codings converged (i.e., were identical across the compared sources). During this process, the adjudicator also flagged any cases that remained ambiguous after review or cases in which the final adjudicated value differed from the original Visible Learning database entry. One example of an ambiguous case was when authors reported multiple meta-analytic results that inform the same interpretation and did not indicate which result should be treated as the primary estimate.

Gold standard The gold standard was subsequently created to resolve these most challenging cases and achieve the highest possible level of accuracy. It was established through a final consensus discussion between the first author, the last author, and John Hattie. We discussed all 67 meta-analyses in which the silver standard value differed from Visible Learning or the adjudicator flagged a need for discussion. In the discussion, we examined all prior values and the full text of the meta-analyses. Again, the gold standard value could differ from all prior extracted values (19 cases). This discussion specifically focused on the set of cases flagged by the silver standard adjudicator, ensuring that the full expert team resolved the most complex and contentious data points. The discussion addressed a spectrum of issues, ranging from high-level conceptual questions (e.g., whether an outcome qualified as 'achievement') to choices about how to handle manuscript errors and minor numerical discrepancies, which occasionally required manual recalculation to verify the correct sample size.

Analytical Procedure to Compare Data Extracted by Human and LLMs

At the end of Phase 3, all extracted values (from each human source and each LLM) and the two reference standards were compiled in a single Excel worksheet with one row per meta-analysis and one column per variable and extraction source, thereby enabling computation of accuracy metrics. The analysis dataset and the R code used to reproduce all reported results are available in the Supplementary Material.

We initiated our analysis by examining the distribution of accuracy metrics across all variables. To address Research Question 1 (How accurately do LLMs extract data?), we calculated the percentage agreement, Intraclass Correlation Coefficient (ICC), Pearson correlation (r), and Mean Absolute Error (MAE) for the LLMs relative to the reference standards. Table 1 summarizes definitions, strengths, and limitations. We interpret percentage agreement using the labels low (<0.60), moderate (0.60 – 0.70), acceptable (0.70 – 0.80), high (0.80 – 0.90), and very strong (>0.90 ; Norouzian, 2021). For ICC(2,1) and correlation, we use poor (<0.50), moderate (0.50 – 0.75), good (0.75 – 0.90), and excellent (>0.90) as general guidelines. For MAE, smaller

Table 1 Metrics to operationalize accuracy

Metric	Definition	Strength	Limitation
Percentage agreement	Proportion of data exactly matching the standard out of all coded fields.	Simple, transparent, task-agnostic.	Inflated by class imbalance; ignores near-misses for numeric values.
ICC(2,1)	Two-way random-effects, absolute-agreement, single-rater intraclass correlation between two coders.	Captures absolute agreement (level + rank); standard for rater reliability.	Assumes approximate normality; sensitive to restricted range and outliers.
Pearson correlation (<i>r</i>)	Linear association between data from two coders.	Detects monotonic alignment; easy to compare across variables.	Insensitive to systematic bias (e.g., constant offsets or scale shifts).
Mean absolute error (MAE)	Mean absolute difference between extracted values rounded to two decimals.	Direct error magnitude and robust to outliers.	Scale-dependent; not comparable across differently scaled variables without normalization.

Note. We had no missing values in the study. If information was missing in the article, it was explicitly coded as NA. We used ICC(2,1) as we considered both coders and studies to be random effects

values indicate higher accuracy. Because MAE is scale-dependent, we contextualize it by the variable's scale range and standard deviation.

Similarly, to address Research Question 2 (How accurately do human coders extract data?), we applied the exact same analytical process to the two human-coded datasets. The original Visible Learning data and the author coding were each compared against the silver and gold standards using the same set of accuracy metrics (percentage agreement, ICC, *r*, and MAE). This allowed for a direct and equivalent comparison of the accuracy of the human coders to that of the LLMs.

To visualize agreement and disagreement patterns across all human and machine coders, we utilized UpSet plots (Lex et al., 2014). An UpSet plot is a powerful tool for visualizing the intersections of multiple sets; in our case, it was used to show the frequency how often combinations of coders (e.g., which LLMs, which humans) agreed or disagreed on specific data points. This allowed us to move beyond simple pairwise agreement to identify complex error patterns, such as errors unique to a single LLM, errors common across all LLMs, or errors shared between a human coder and a specific AI. It provides information about the pattern and could indicate how often humans agree with humans and how often humans agree with LLMs.

Furthermore, we used Bland-Altman plots to provide a visual analysis of the (dis) agreements, enabling us to evaluate for: (1) systematic bias, indicated by whether the mean difference deviates from zero, and (2) limits of agreement, which define the range within which most differences between two methods are expected to fall.

This graphical analysis helped determine whether one method consistently over- or underestimated values relative to another, and whether the magnitude of the error was related to the magnitude of the effect size.

Results

Descriptives

The distributions of effect sizes, the number of studies, and the total number of participants are displayed in Fig. 2. The violin plots show similar distributions across LLMs (green), human raters (blue), and both standards (silver and gold). The means for effect sizes were tightly clustered ($M = 0.50$ – 0.51 , $SD = 0.33$ – 0.35 ; $\min = -0.70$ to -0.43 ; $\max = 1.58$). For the number of studies, means ranged from 41.54 to 46.68 ($SD = 38.75$ – 59.33 ; $\min = 5$; $\max = 271$ – 594). Total participants showed wider dispersion ($M = 32,589$ – $76,863$, $SD = 143,264$ – $260,455$; $\min = 76$ – 651 ; $\max = 1,100,000$ – $1,663,662$).

While the distributions were very similar, differences in at least one data point existed in 101 of 156 meta-analyses. All of these meta-analyses were adjudicated by

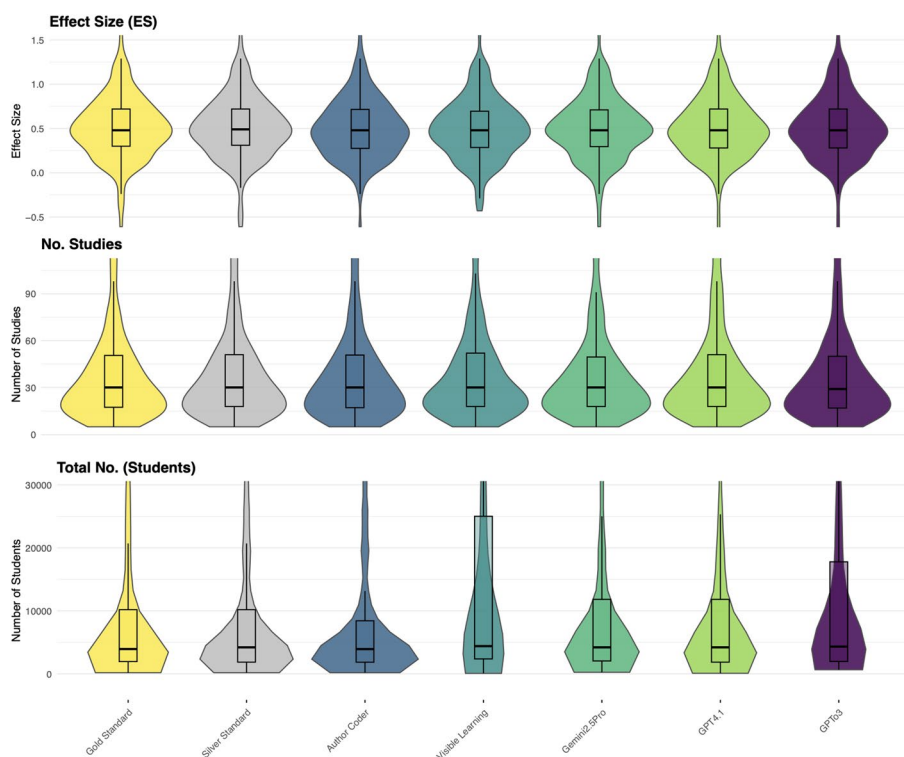


Fig. 2 Violin plots showing the distribution for gold/silver standard, human coder, and the LLMs. Note Violin plots showing the distribution for gold/silver standard, human coder, and the LLMs

one author to establish the silver standard. From those adjudicated meta-analyses, 67 meta-analyses (42.9% of the full sample) needed further discussion to reach gold standard consensus because residual ambiguity remained after first adjudication or because the adjudicated value still diverged from the Visible Learning entry. The adjudication effort concentrated on the total number of participants. Most disagreements in N occurred when participant counts were distributed across multiple tables or subgroups and had to be summed, leading to small discrepancies due to calculation errors. The remaining disagreements were largely driven by ambiguous reporting (for example, conflicting values across abstract and results sections or multiple plausible ‘overall’ effects). Hallucinations in the strict sense were rare; when LLMs erred, they typically selected a value reported in the manuscript but did not match the codebook target, such as a subgroup estimate rather than the primary overall synthesis.

Figure 3 illustrates the number of agreements for each combination of coders across the 468 data points. The highest line on the left indicates that in 280 cases (59.8% of the 468 comparison points), human, LLM coder, and both standards agree. The next most frequent pattern is a single-coder human deviation in the Visible Learning data (57 cases; 12.2%). In line with that, in 27 cases (5.8%), all coders, except Visible Learning, agreed. Interestingly, we found only 16 cases (3.4%) in which both humans agreed with the standards, and none of the LLMs made an accurate extraction. There are 14 cases (3.0%) where one human (either from Visible Learning or the author coding) agreed with all three LLMs, and the adjudication still yielded different values. Note, that the absence of cases where the three LLMs and both humans agreed but the silver/gold standard disagreed is due to our procedure, as we did not adjudicate these values and assumed they were accurate (see limitations for further discussion).

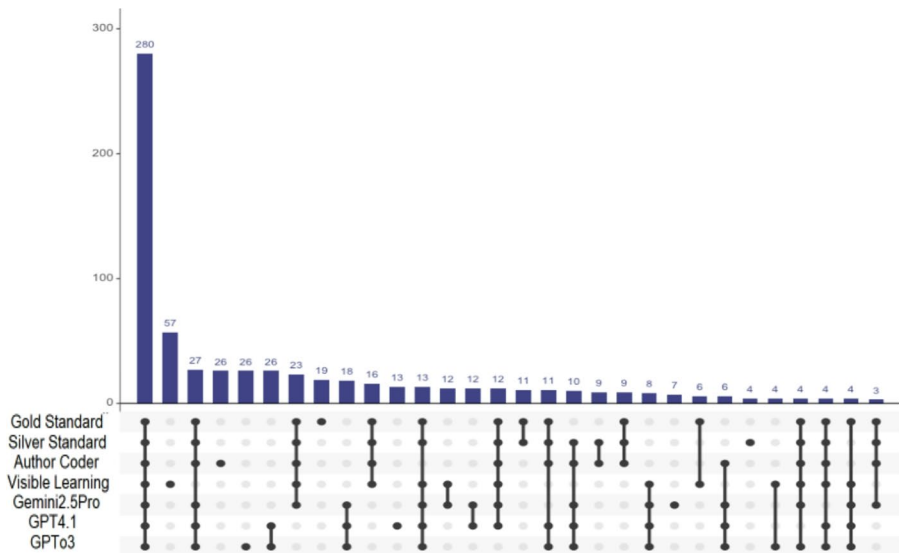


Fig. 3 UpSet plot showing which combination of coders agrees with another and the gold and silver standard

An analysis of specific error types reveals distinct performance profiles for the human and LLM coders. We defined hallucinations (following Gartlehner et al. 2024b) as instances in which a coder reported a value when the silver/gold standard indicated that no such information was present in the source text. Omissions as the inverse, where a coder failed to extract a value that was present. The results show that hallucinations were a rare error type across all sources. Both expert human coders and LLMs produced very few fabricated data points, with the number of hallucinations ranging from 3 to 4 cases among the 55 meta-analyses for which that information was missing in the full text. This indicates a high degree of reliability in avoiding the creation of non-existent data for all coders. In contrast, omission errors were a more frequent challenge (i.e., coding information as missing when it was included in the manuscript; Gartlehner et al. 2024b). The independent author coding demonstrated the lowest rate of omissions, failing to extract an existing value in only 12 instances. The state-of-the-art LLMs (Gemini 2.5 Pro and GPT 4.1) showed moderate omission rates, with 21 and 22 errors, respectively, performing better on this metric than the Visible Learning data, which had 31 omissions. The older model, GPT o3, was most prone to this error, with the highest omission count at 36. All omissions were in the number of included students.

Analyses RQ1 and RQ2: Accuracy of LLM-Extracted compared to Human-Extracted Data

The accuracy of the LLM and human coders in relation to the adjudicated standards is detailed in Table 2 (see Supplementary Material A for tables regarding each variable). The results indicate a high degree of accuracy for the LLMs when compared against the gold standard. Across the three models, percentage agreement ranged from 77% to 81%, while the Intraclass Correlation Coefficients (ICC) were excellent, ranging from 0.96 to 0.97. Their performance was similarly strong against the silver standard, with agreements of 78% to 84% and ICCs ranging from 0.94 to 0.96. The high level of agreement among the LLMs themselves (82%–90% agreement; ICCs of 0.95–0.97) further indicates that their performance was consistent and converged on the correct values rather than achieving sporadic matches with the standards.

When benchmarked against human coders, the LLMs demonstrated performance comparable to that of expert-level extraction (see Table 2). The independent author

Table 2 Accuracy matrix of the LLM and human coders with the standards. Upper triangle shows percentage agreement, and lower triangle ICC (2,1)

	Gold Standard	Silver Standard	Author coding	Visible Learning	Gemini 2.5Pro	GPT 4.1	GPT o3
Gold Standard	—	93%	86%	80%	81%	78%	77%
Silver Standard	0.98	—	91%	79%	84%	80%	78%
Author coding	0.95	0.98	—	76%	82%	78%	77%
Visible Learning	0.81	0.78	0.75	—	79%	71%	71%
Gemini2.5Pro	0.96	0.95	0.92	0.81	—	87%	82%
GPT 4.1	0.97	0.96	0.94	0.79	0.97	—	90%
GPT o3	0.96	0.94	0.92	0.78	0.95	0.97	—

coding achieved 86% agreement ($ICC = 0.95$) with the gold standard and 91% agreement ($ICC = 0.98$) with the silver standard. The LLMs' accuracy fell within a narrow band of this high-quality human performance. In contrast, the original Visible Learning data exhibited lower agreement with both the gold standard (80%, $ICC = 0.81$) and the silver standard (79%, $ICC = 0.78$). A visual inspection of the disagreements using Bland-Altman plots (Supplement A) revealing no evidence of systematic bias among any of the human or LLM coders; that is, no coder consistently over- or underestimated the extracted values. Further, it shows that the reduced ICC for the Visible Learning data is largely attributable to a single extraction error in the meta-analyses from Gunter and Shao (2016), where the number of effect sizes (594) was incorrectly coded as the number of studies (gold standard = 9). Additionally, the slightly different violin plots for the number of included students in the Visible Learning GPT o3 data may be due to disagreements over single outliers in the meta-analyses. Pahlke and colleagues (2014) reported in the abstract that they included approximately 1.6 million students, but later provided the correct number (1,663,662), leading to a significant discrepancy.

Discussion

As an applied field, educational psychology depends on comprehensive research syntheses, such as SOMA, to structure and integrate extensive bodies of evidence, thereby facilitating evidence-based decision-making. SOMAs provide researchers, practitioners, parents, and policymakers with a systematic overview of central research domains. The present study's primary contribution to conducting SOMAs in a rigorous and efficient manner is to provide an empirical foundation for using LLMs to accelerate data extraction from meta-analyses.

The focus of our discussion is to provide practical guidance for researchers on the responsible integration of LLMs into data extraction processes while maintaining meaningful human control. Such methodological rigor is essential given the influence of SOMAs on research, policy, and practice, as exemplified by the Visible Learning project. Before detailing specific use cases, we discuss our findings in the context of the initial research questions and the literature on automated data extraction using LLMs.

Research Question 1: LLMs Extracted Data for SOMA were Highly Accurate

Our results demonstrate that LLMs can extract the results from meta-analytic data with excellent Intraclass Correlation Coefficients (ICC ranging between 0.94 and 0.97). Such high $ICCs$ indicate that even in the cases of disagreement between the gold/silver standard and the LLMs, these disagreements were minor and did not change the rank order of results. Looking at the distributions of coded variables and errors in detail reinforced this interpretation. Errors included minor calculation discrepancies when the models summed participant counts across multiple included primary studies in the meta-analysis, or plausible miscoding due to ambiguity in the articles. The most severe errors were three cases in which the sign of the effect size

was wrong (which occurred in human coding to a similar extent). Nevertheless, up to 23% of extractions were not fully accurate, indicating a clear margin for improvement that future methodological and workflow advances should aim to reduce.

The observed accuracy suggests that findings from other contexts are transferable to SOMAs from educational meta-analyses, as they align with a growing body of evidence from other contexts demonstrating that LLMs can extract data with high accuracy (e.g., Konet et al., 2024; Khan et al., 2025; Reason et al., 2024), low error rates (Jensen et al., 2025; Khan et al., 2025), and no systematic bias (Jansen et al., 2025). Placing the error rates of LLMs in perspective, it is worth noting that they are lower than those of humans attempting to replicate meta-analyses (37% error rates; Göttsche et al., 2007). The high mean accuracy in our study surpasses that reported in earlier investigations (e.g., Gue et al., 2024; Mahmoudi et al., 2024; Sun et al., 2024). We attribute the difference to the relatively optimal conditions in our study, combining the use of highly capable LLMs, a focus on clearly reported variables, and carefully optimized prompts. We discuss how these rather optimal conditions constrain the generalizability of interpretation to both research questions in the Limitations section below.

Research Question 2: LLMs Approach the Extraction Accuracy of Expert Humans

Our findings indicate that the LLMs achieved the data-extraction accuracy of expert human coders across all metrics, including agreement, ICC, systematic bias, and hallucinations, for the investigated extraction task. The expert human coder achieved an average agreement of 83% with the gold standard. Even the silver standard only agreed with the gold standard in 93% of cases. These agreement values are rather high compared to prior studies, which showed 37% error rates among experts trying to replicate meta-analytic findings (Göttsche et al., 2007; see the review from Schmidt et al., 2025 for more examples). We attribute the high accuracy in our study to both substantial coder expertise and a task likely at the easier end of the data-extraction spectrum.

Investigating disagreements in detail during our adjudication process revealed that most errors stemmed from ambiguity within the source meta-analyses, and all human extraction was plausible. Issues such as complex reporting, the need to aggregate data sample descriptions across multiple tables were common sources of discrepancy. We were surprised that even after an initial expert adjudication building the silver standard, 67 cases (43% of the 155 adjudicated cases) were flagged as benefiting from a final discussion, highlighting that even expert-driven adjudication cannot eliminate uncertainty in evidence synthesis. The discussion with John Hattie yielded different conclusions than all prior values in 19 cases. These cases illustrate a rather optimal adjudication scenario in which the original SOMA's first author was available for direct discussion, enabling ambiguities in reporting to be resolved at their source; in principle, the most definitive resolution would involve direct consultation with the original authors of the coded studies.

The finding that the best-performing LLMs operate within a very narrow margin relative to human experts necessitates a decisive moment in evidence synthesis regarding the standards that human and LLM coders must achieve. The question is

whether we should apply a single or dual standard to data extracted by humans and LLMs. This decision has profound implications. Suppose we adopt the same single standard for human and AI accuracy. In that case, LLM's extraction accuracy will be accurate enough. Combined with LLMs' extreme efficiency (<6 min vs. ~390 h), data extraction might become a task performed almost exclusively by LLMs (see Thomas et al. 2025a, b for a contrasting position statement). Conversely, if we decide on a dual standard requiring LLMs to be more accurate than humans to keep humans in the loop, because humans ultimately take responsibility for the final data (Flemyng et al., 2025), the research community would need to define, measure, and justify a super-human level of accuracy that we ourselves cannot consistently achieve.

To move forward respecting these tensions, it is necessary to go beyond the current status, where the evaluation of data extraction is largely reduced to reporting inter-coder agreement between humans. Agreement metrics indicate whether outcomes are similar, but they provide limited information about whether researchers using LLMs can understand the extraction process, intervene when errors occur, and take responsibility for the resulting data. We therefore argue that evaluations should move beyond outcome similarity and explicitly examine whether meaningful human control is maintained when LLMs are used for data extraction. In practice, this requires workflows in which researchers can inspect, interpret, and correct LLM outputs and in which the extraction process itself is transparent. In the following sections, we present concrete use cases illustrating how LLMs can support data extraction while preserving meaningful human control.

Practical Applications: Questions and Answers

Our results suggest that for SOMA from educational meta-analyses, LLMs can already be valuable for extracting effect sizes, the number of included studies, and the number of included participants from meta-analyses. We interpret the findings this way because prior work has shown that LLMs operating at comparable levels of accuracy can be leveraged to substantially accelerate evidence synthesis without increasing overall error rates (e.g., Cao et al., 2025; Gartlehner et al., 2025). Hence, the following section address most pressing question about practical applications.

But before turning to specific use cases, we endorse the recommendations outlined in the living guidelines on the responsible use of generative AI in research (European Commission, 2024), the recommendations from the Responsible use of AI in evidence Synthesis group, (RAISE; Thomas et al. 2025a, b,) and the principles for protecting scientific integrity in an age of generative AI (Blau et al., 2024). In line with these guidelines, comprehensive reporting of all AI-assisted steps throughout the review process is essential for maintaining the integrity and robustness of scientific inquiry. This process requires disclosing specific details, such as the used LLMs version and settings, the prompt engineering process and final prompt versions, as well the performance metrics measured during the pilot evaluation phase. Such transparency is essential for enabling replication, building trust in the validity of SOMA findings, and fostering the responsible integration of AI in research synthesis.

Where is the Shown LLMs' Accuracy "Good Enough" for Educational Meta-analysts?

Quality Assurance and Error Detection Our results suggest that an LLM can serve as a quality assurance tool when conducting SOMA (see Jensen et al., 2025; Mahuli et al., 2023 for aligning propositions). The UpSet plot (Fig. 3) revealed 27 instances in which a human coder was the sole outlier relative to a consensus among all other human and machine coders. The present study provides a concrete example of this function, as LLM-generated extractions repeatedly prompted discussion and led to improvements in the Visible Learning database. In this configuration, meaningful human control is preserved because LLMs only suggest possible errors, while humans retain full authority over verification, adjudication, and implementation of changes. As a result, researchers remain accountable for both the extraction process and the synthesized evidence.

Providing Extraction Assistance for Humans LLMs can act as extraction assistance, as LLMs' extraction in similar quality has been shown to provide useful assistance, accelerating the data extraction process, while keeping error rates constant (Gartlehner et al. 2025; Wang et al. 2025b). For educational researchers conducting SOMA, this support can be provided through interfaces such as those proposed by Schroeder and colleagues (2025) or through commercial systems like Google's NotebookLM or Elicit (Bianchi et al., 2025), which present a concrete proposition for each extracted data point and link it to the exact location in the source PDF (These systems were published after the data collection for the present study was completed and therefore could not be used here). In this workflow, meaningful human control is preserved because every LLM-generated extraction is explicitly verified by a human coder. This verification allows researchers to assess both the correctness of the extracted values and the reliability of the LLM's extraction behavior.

Be a Second Independent Extraction Agent Adding to Primary Human Coding Educational researchers conducting SOMA can operationalize the principle that rigorous data extraction requires at least two independent codings by pairing an LLM with a human coder (Helms Andersen et al., 2025; Jensen et al., 2025; Reason et al., 2024; Wang et al., 2025a, b). Any disagreements between the two should be resolved through adjudication by an additional human who was not involved in the initial coding and who follows a blinded adjudication procedure to reduce bias (Cao et al., 2025). The first coder needs to be a human researcher, preferably part of the author team, so that every extracted value in the SOMA has undergone at least one human verification.

Where are LLMs Not Yet Acceptable for Data Extraction?

An intuitive use case is to enable independent data extraction by multiple LLMs, with humans adjudicating only when LLMs disagree. We argue that it is not an acceptable use case because humans do not have the possibility to verify every outcome, and so meaningful human control cannot be established in this way. In such a setup,

agreement between LLMs is implicitly treated as correctness, even though our results show that unanimous agreement can still yield incorrect extractions. As a consequence, responsibility for large parts of the dataset would be effectively delegated to the system rather than retained by human researchers.

A common practice in human coding is to establish high agreement on a subset of studies and subsequently proceed with single coding. Transferred to LLM-based data extraction, this would involve demonstrating high agreement between humans and LLMs on a subset of studies and then allowing the remaining data to be extracted exclusively by the LLM. We argue that this approach is insufficient for meaningful human control because humans would no longer have the opportunity to verify every extracted outcome.

Another edge-case use of this approach is having one human and one LLM perform independent data extractions, then having the LLM adjudicate disagreements. In this setup, humans can no longer influence the final resolution of conflicting extractions, implying that human control over the end product is lost.

When LLMs and Human Coders Agree, is the Data Correct?

A tempting assumption is that when a human coder and an LLM agree, the data point can be considered validated and accepted without further review. Our findings urge caution with this approach. We identified cases in which an LLM and a human coder agreed, yet the adjudicated gold standard showed their consensus was incorrect. This indicates that human-LLM agreement, while a positive signal, is not an infallible guarantee of accuracy. Complicating the issue, we also found cases in which both humans agreed, yet the adjudication showed they were wrong. Thus, using an LLM for validation is useful, but only as useful as adding another human layer of data extraction.

When is a Human–LLM Hybrid Workflow Preferable to Fully Manual Coding?

While the empirical basis for various specialized use cases remains under development, our results and the emerging literature indicate that a human-LLM hybrid workflow is superior to a human-only approach under identical resource constraints. Considering that traditional systematic reviews can require up to \$100,000 in financial resources and take over a year to complete, the marginal cost of integrating LLM assistance is negligible. Setting up and executing automated extraction requires only a few hours compared to the approximately 390 h of specialized human labor needed for manual extraction in a corpus of this size, and it costs only 50\$ compared to the much higher cost for the specialized human labor.

Limitations and Future Research

The present evaluation targets a comparatively favorable (“best-case”) extraction scenario for both humans and LLMs, which can overstate typical data-extraction accuracy. We focused on three standardized numeric variables that were retrieved

(not inferred) from the manuscript text, evaluated state-of-the-art proprietary LLMs (i.e., in the moment of writing), and compared them against expert human coders as the “status quo” benchmark. The following limitations explain how the specific context of our study, the nature of the human and LLM coders, and the rapid evolution of the technology constrain interpretation.

Limitations Due to the Chosen Context

Our study deliberately focused on three core numeric variables (effect size, number of studies, and number of participants) extracted from educational meta-analyses in the Visible Learning database. This is a comparatively well-structured extraction setting because these variables are central to meta-analytic reporting and are often repeated across tables, abstracts, and results sections. Accordingly, our accuracy estimates should be interpreted as upper bounds (best cases) for both humans and LLMs and should not be generalized to extraction tasks that require construct interpretation, judgment, or classification. Extraction performance is known to be variable-dependent (Jansen et al., 2025), and our results may not extend to less uniformly reported variables or to interpretive coding tasks (e.g., moderator categories, operationalizations of achievement, intervention features, risk of bias in meta-analyses, publication bias).

However, this limitation is less pronounced for our second research question (RQ2), which concerns the comparative accuracy of LLMs and human experts. The use of clearly defined, fundamental variables likely increased the accuracy of both human and machine coders, creating a high-performance environment for the comparison. Given that human data extraction in other contexts has been shown to have error rates as high as 50% (Mathes et al., 2017), our results for human accuracy likely represent a best-case scenario. Therefore, the finding that LLMs approach this expert benchmark should be interpreted as valid within this narrow, best-case extraction scope, rather than as evidence that LLMs match humans for educational review coding in general.

Limitations Due to the Human Coders

Our findings regarding human coding accuracy are subject to limitations of generalizability. A small team of researchers conducted Visible Learning and the author coding. Furthermore, these individuals were not representative of all potential coders but were experts with multiple years of experience in coding educational meta-analyses (e.g., Fleckenstein et al., 2023; Jansen et al., 2022; Möller et al., 2020). In contrast, many educational reviews rely on teams with mixed experience levels (including graduate assistants or novice coders), hence our human benchmark should be read as an upper-bound estimate of manual extraction performance, not as a field-wide status quo. Additionally, the similarity in the distribution of extracted variables that we observed between LLMs and humans may not generalize to scenarios involving novice or less-experienced coders, who might exhibit different error rates and patterns.

Limitations Due to Our Choice of the Human Comparison Standard

The creation of the silver and gold standards is subject to two methodological limitations. First, the adjudication process was conducted by the study authors, who were also involved in the initial author coding. While it is standard practice in meta-analytic research for the author team to resolve discrepancies, this design is not fully blinded and introduces a potential for confirmation bias, where adjudicators might unintentionally favor their original ratings. An independent validation such as done by Cao and colleagues (2025) would be required to eliminate this possibility.

Second, our procedure assumed that any data point on which all initial coders (both human and LLM) were in unanimous agreement was correct, and these cases were therefore not subjected to adjudication. This approach is based on the common convention that agreement between two independent coders is sufficient to establish accuracy. On the one hand, the assumption might be incorrect, as we observed that in 19 cases (4.1%), the gold standard mentioned in the discussion differed from all previous coded values. On the other hand, these consensus cases were typically those where the data were clearly presented (e.g., all values reported in the abstract), and when four coders agreed, the accuracy was already 96%. For future studies, it would be especially interesting to determine how many human coders or LLMs, when extracting data across multiple runs or with different prompts, need to agree to ensure the data is validated. Based on our data, one might assume the turning point is around four coders, but other studies show that even with eight LLMs, incorrect extractions still occur (Jansen et al., 2025).

Limitations Due to the LLM Usage

A key methodological choice that limits the direct replicability of our findings was our intensive and iterative prompt engineering process. While we provide all final prompts for transparency, this highly optimized process is not standardized and is tied to our specific data and expertise. Consequently, the accuracy estimates may not readily generalize to other teams using different or less refined prompting strategies. Yet, studies that used more naive prompts (e.g., Konet et al., 2024; Reason et al., 2024) still reported high accuracy rates.

Our prompt was iteratively refined on a pilot set, which introduces a potential risk of overfitting the extraction rules to a specific subset of meta-analyses. To mitigate this, the pilot set ($n=10$) was kept fully disjoint from the final test set ($n=156$), which was sampled only after the prompt had been finalized, and we provide the complete pilot materials and optimization traces in the Supplementary Material for transparency and reproducibility. Nevertheless, in a broader sense our domain familiarity with educational meta-analyses inevitably shaped the prompt design, so the resulting performance estimates should be interpreted as reflecting a well-calibrated, expert-informed workflow rather than a universally optimal prompt that would generalize unchanged to other corpora or review teams.

Our decision to use proprietary, black-box LLMs rather than traditional AI systems raises practical and ethical concerns that warrant explicit acknowledgment. Reliance on proprietary models reduce reproducibility and long-term accessibility, as

models may be updated without notice, and usage costs can change over time, potentially affecting the stability and availability of results. In addition, LLMs operate as largely opaque systems, complicating interpretability and error diagnosis compared with rule-based or task-specific algorithms and reducing transparency in the research process (Demszky et al., 2023).

The use of LLMs carries sustainability costs (see Thomas et al. 2025a, b, for discussion); specifically, the computational energy required for model inference (e.g., Elsworth et al., 2025) is higher than that of traditional algorithms. Infrastructure-aware benchmarking (Jegham et al., 2025) indicates that the energy required for a single long-form query (10,000 input and 1,500 output tokens) ranges from approximately 4.83 Wh for models such as GPT 4.1 to over 29 Wh for more energy-intensive reasoning systems. In the specific context of this study, processing 156 meta-analyses across three LLMs would consume an estimated 1.3 to 2 kWh of electricity and approximately 5.3 L of evaporated freshwater. When interpreting these impacts, one must consider the comparative environmental footprint of manual labor; the approximately 390 h of specialized human effort required to complete the same task would involve significantly higher workstation electricity consumption and personal water needs.

Our selection of LLMs also constitutes a limitation. In line with our goal of assessing the upper bound of current data extraction capabilities, we deliberately chose proprietary, state-of-the-art models that were top performers at the time of our study. This decision, however, means our findings do not represent the full landscape of available technologies and are not generalizable to other types of models, particularly open-source alternatives, which may have different performance characteristics. Therefore, our results should be interpreted as a benchmark of what was achievable with the most powerful models at that specific point in time, rather than an evaluation of all LLMs. There is the concern of “train/test contamination” or data leakage when using LLM benchmarks (Wang et al. 2025b), given the extensive web scraping used to train these models (Woelfle et al., 2024). Thus, there is a risk that LLMs achieve their performance not because they extracted data from the studies’ PDFs but because they searched for the information within their training data. The Visible Learning database is likely to be included in the training data, which could lead to an overestimation of its true extraction capabilities on novel data. While we cannot definitively quantify this risk with our current design, two factors suggest its influence is likely limited. First, our prompt architecture strongly grounds the LLMs in the provided text of each meta-analysis, making it the most salient source of information for the task. Second, and more critically, our results include numerous instances in which the LLMs’ extractions correctly aligned with our gold and silver standards, even when those standards diverged from the original values in the Visible Learning database. This pattern of agreement suggests that the LLMs were performing genuine extraction from the source documents rather than merely regurgitating pre-existing data. Nevertheless, we acknowledge this limitation, and only prospective replications using novel, non-public human consensus datasets can conclusively eliminate the risk of train/test contamination.

Limitations due to the Fast-development of LLMs

In fast-moving research fields, research findings are inevitably time-sensitive, and our results are no exception. The landscape of LLM capabilities is shifting so quickly that replicating our evaluation study, even within a short timeframe, could yield notably different outcomes. Nevertheless, we believe our work remains valuable given the assumption that LLM performance will continue to improve.

Conclusion

The time-consuming nature of data extraction hinders the merits of SOMA. Our study demonstrates that for a specific data extraction task LLMs performed on par with expert humans. Thereby, we provide an empirical foundation for responsible use of LLMs for data extraction.

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Authors' Contributions Thorben Jansen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing. Lucas Liebenow: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. Jonathan Schaller: Methodology, Software, Validation John Hattie: Supervision, Investigation, Data curation, Writing – original draft, Writing – review & editing. Jens Möller: Supervision, Investigation, Data curation, Writing – original draft, Writing – review & editing. All authors have read and agreed to the published version of the manuscript.

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Data Availability Data analyzed here are available in the supplementary material and under : <https://osf.io/q5n7f>

Declarations

Ethical Approval Not applicable because no human and/or animal participants involved.

Competing Interests No competing interest (always applicable and includes interests of a financial or personal nature).

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